

AN **INDIAN** ADAPTATION

Third Edition

Supply Chain Management

Nada R. Sanders
Sidhartha S. Padhi

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SUBJECT: SUPPLY CHAIN MANAGEMENT STRATEGY

CHAPTER: 05

CHAPTER TITLE: OPERATIONS MANAGEMENT

REFERENCE BOOK:

SUPPLY CHAIN MANAGEMENT (AN INDIAN ADAPTATION), N R SANDERS & S S PADHI, WILEY, 3RD ED. (2023)

KEY CONCEPTS

Operations management (OM)	Operations management (OM)—or “operations”—is the business function responsible for producing a company’s goods and services in an efficient and cost-effective way. It is the function responsible for transforming a company’s inputs, obtained by the sourcing function, into finished goods and services, which marketing sells to final customers. Without operations, there would be no products to sell, whether tangible goods or services.
OM FUNCTION	OM is responsible for planning, organizing, and managing all the resources needed to produce a company’s goods and services. This includes people, equipment, technology, materials, and information. It is a core function of every company, whether large or small, service or manufacturing, profit or nonprofit.
OM DECISIONS	OM is complex as there are many different types of decisions that are involved in transforming inputs into finished products. For operations to be able to design and produce the products customers want, customer preferences such as likes and dislikes have to be translated into tangible product characteristics. It is then up to operations to decide on the exact type of process needed to produce the desired product. This includes equipment, level of technology, automation, skill levels, job requirements of workers, and capacity of facilities.
PRODUCT DESIGN	Product design is the process of specifying the exact features and characteristics of a company’s product. All of a product’s features are determined by product design.

KEY CONCEPTS

PROCESS DESIGN	Process design is the design of the production process needed to produce the desired product.
DESIGN OF SERVICES	Product design in the service industry has an added complication as the product is intangible and there is a high degree of customer interaction. Service design requires the design of the entire service concept. As with a tangible product, the service concept is based on satisfying customer needs.
BREAK-EVEN ANALYSIS	Break-even analysis is a tool that can be used to evaluate the success of a new product at the product screening stage.
PRODUCT LAYOUTS	Product layouts arrange resources in sequence so the product can be made as efficiently as possible. This type of layout is used when producing a large volume of one standardized product and there is repetition of the process.

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SECTION 1

OPERATIONS MANAGEMENT (OM)

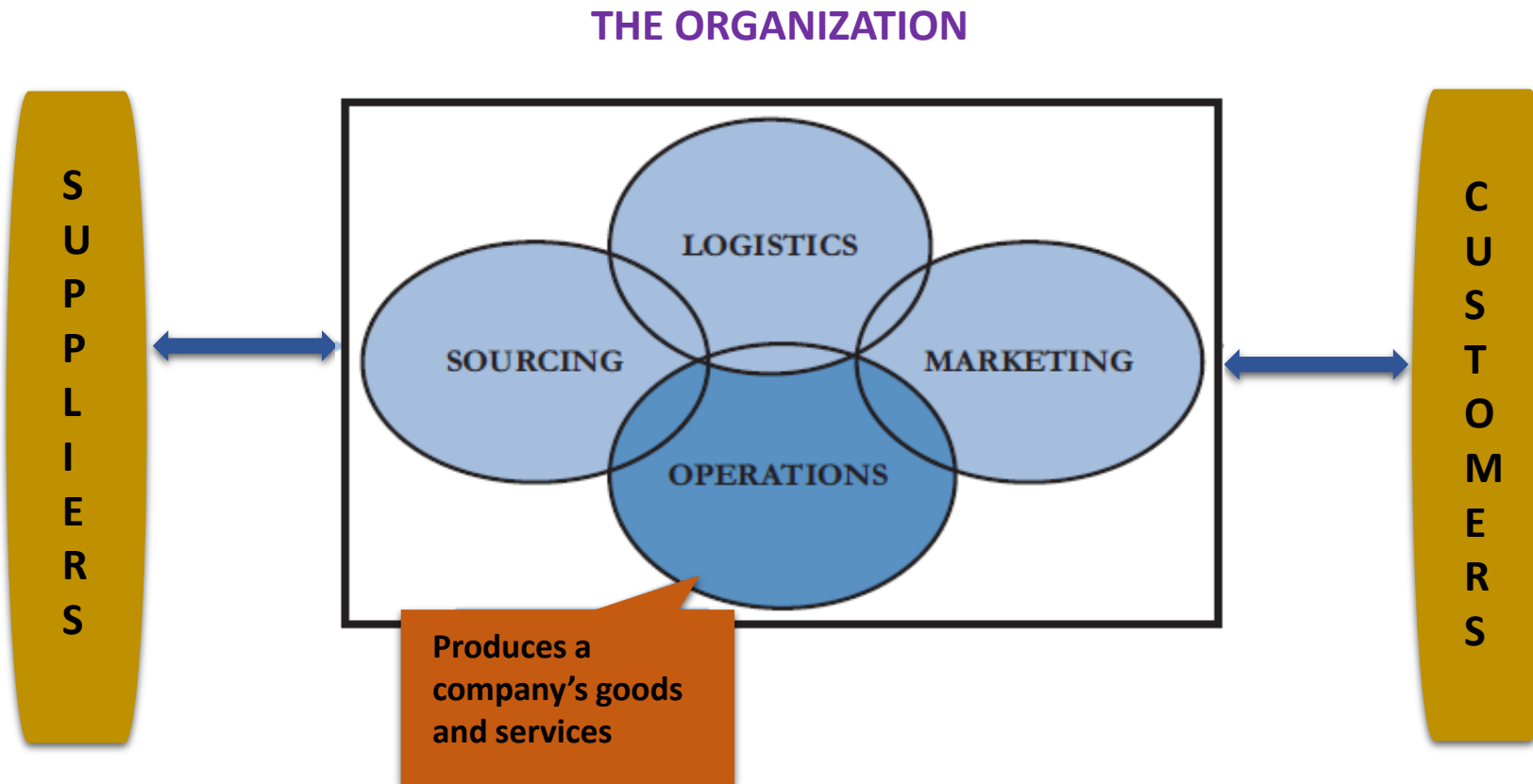
OM FUNCTION

- *Operations management (OM) or “operations”* is the business function responsible for producing a company's goods and services in an efficient and cost-effective way.
- It is the function responsible for transforming a company's inputs, obtained by the sourcing function, into *finished goods and services*, which marketing sells to final customers.
- Without operations, there would be no products to sell, whether tangible goods or services.
- The functional position of operations in the organization is highlighted in Figure 1.
- OM is responsible for *planning, organizing, and managing* all the resources needed to produce a company's goods and services. This includes *people, equipment, technology, materials, and information*. It is a core function of every company, whether large or small, service or manufacturing, profit or nonprofit.
- Consider Gap Inc., one of the largest specialty retailers. Gap's marketing function identifies characteristics of products desired by customers for each of its three brands, Banana Republic, Gap, and Old Navy.
- However, it is OM that must design and produce the merchandise, from T-shirts to jeans. It must also ensure that the right products in terms of variety, colors, and styles are available in the right quantities at various retail locations across the globe.

OPERATIONS MANAGEMENT (OM)

- OM serves a *transformation role* in the organization by converting a company's inputs into finished goods and services, shown in Figure- 2. These inputs include human resources, such as workers, staff, and managers; facilities and processes, such as buildings and equipment; it also includes materials, technology, and information.

Figure-1: The operations function



OPERATIONS MANAGEMENT (OM)

Figure- 2: The transformation role of OM



- Outputs are the goods and services a company produces. At a manufacturing plant the transformation is the physical change of raw materials into products, such as transforming leather and rubber into sneakers, denim into jeans, or plastic into toys.
- At a hospital it is organizing resources, such as doctors, medical equipment, and medications, to transform sick patients into healthy people.
- Consider the example of *Ryanair*. Like traditional airlines, *Ryanair* transports passengers and their luggage from one location to another. However, the transformation process at *Ryanair* is organized very differently from traditional airlines, by involving the customer in aspects of the transformation process itself.

OPERATIONS MANAGEMENT (OM)

OM DECISIONS

- OM is complex as there are many different types of decisions that are involved in transforming inputs into finished products. Some of these are listed in Figure-3. For operations to be able to design and produce the products customers want, customer preferences such as likes and dislikes have to be translated into tangible product characteristics. It is then up to operations to decide on the exact type of process needed to produce the desired product. This includes equipment, level of technology, automation, skill levels, job requirements of workers, and capacity of facilities.
- Consider your university and the transformation role it performs in educating students. Think about the location of the university, size and proximity of classrooms; consider the course offerings, capabilities of faculty and staff, and schedules provided; think about the resources from cafeterias and conference rooms to computers and athletic facilities. These are all examples of OM decisions designed to create a transformation process.

Figure-3: OM decisions

OM Decision	What it means
Product design	Designing unique product features
Process design	Creating the production process to produce the designed products
Quality management	Establishing and implementing quality standards
Inventory management	Decisions on amounts of inventory to carry and when to order
Facility layout	Deciding the physical layout of the production facility
Facility location	Deciding on the best location for facilities
Scheduling	Creating schedules for workers, machines and facilities

OPERATIONS MANAGEMENT (OM)

MANUFACTURING VERSUS SERVICE OPERATIONS

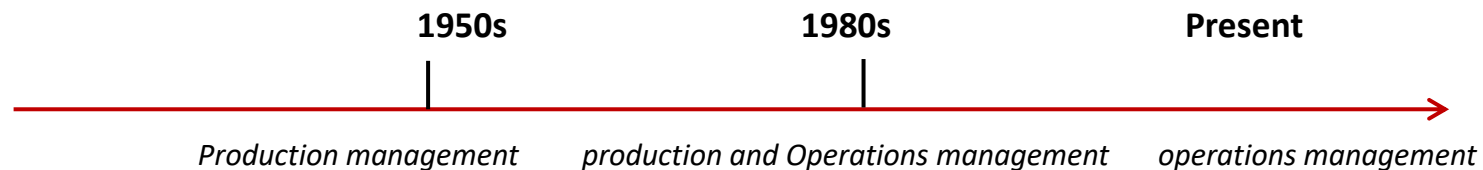
- *Manufacturing and service organizations* each pose unique challenges for the operations function. There are two primary distinctions between them.
- First, *manufacturing organizations* produce physical, tangible, goods that can be stored in inventory before they are needed. By contrast, *service organizations* produce intangible products that cannot be produced ahead of time.
- Second, in manufacturing organizations most customers have no direct contact with the operation.
- For example, *a customer buying a bottle of Pepsi never sees the bottling factory*. However, in service organizations the customer is typically present during the creation of the service, such as at hospitals, colleges, theaters, and barber shops.
- The differences between *manufacturing and service organizations* are not as clear-cut as they might appear, and there is much overlap between them. Most manufacturers provide services as part of their business, and many service firms manufacture physical goods they use during service delivery.
- For example, a manufacturer of furniture may also provide shipment of goods and assembly of furniture. A barber shop may sell its own line of hair care products. Even in pure service companies some segments of the operation may have high customer contact while others have low customer contact.

OPERATIONS MANAGEMENT (OM)

EVOLUTION OF THE OPERATIONS FUNCTION

- OM has significantly evolved and changed over time, as shown in Figure- 4. As a result of global business trends and changes in competitive requirements, operations has evolved from being a **'back room'** production function to taking on a strategic role and becoming a tool for competitive advantage. This is also reflected in the change of its name, which was formerly **'production management,'** indicating that it was just about **'producing.'** The term then evolved into **'production & operations management'** reflecting a broader organizational focus. The function was ultimately coined **'operations management,'** reflecting a broad and strategic organizational reach.

Figure-4: The evolution of OM



- Prior to the 1950s, OM served merely a **production role in the organization**. The key opportunities at the time were in marketing, to develop markets for new products, and in finance, to support this growth. Operations was seen more as a 'back room' function that simply 'made things.'
- Although operations began taking on a broader organizational role in **the 1950s and 1960s**, its role changed abruptly in the **1980s**.
- Today's global competition and focus on **SCM** have placed even greater importance on **operations**. Organizations within the supply chain network must operate at the highest level of efficiency and utilize the most current operations methods. In fact, many inroads that are made in competitive markets come from innovations in operations, such as Tata Motors producing less expensive vehicles due to new production techniques.

OPERATIONS MANAGEMENT (OM)

IMPACT ON THE ORGANIZATION

- **Operations** play a critical role in the organization, as without OM there would be no product to sell. However, operations cannot work in isolation from other business functions. For example, operations must work with marketing to understand and design the exact products customers want and to create the production processes that efficiently produce these products.
- **Marketing**, on the other hand, must understand operations' capabilities, such as the types of products it can, and cannot, produce. Otherwise, the company may find itself marketing products the company isn't able to deliver.
- Consider **a furniture manufacturer in North Carolina** where marketing pushed for the selling of laminated versus customized furniture, only to find that the operations requirements were completely different for one versus the other. Operations must also work with sourcing to understand material availability, such as sources of supply quality standards, and availability of materials. Sourcing, in turn, must understand operational production requirements in order to develop and manage an appropriate supply base.

OPERATIONS MANAGEMENT (OM)

IMPACT ON SCM

- Each company depends on other members of its supply chain to be able to deliver the right products to its customers in a timely and cost-effective manner.
- In the *upstream part* of its supply chain, a company depends on its suppliers for the delivery of raw materials and components in time to meet production needs. If deliveries of these materials are late, or are of poor quality, production will be delayed, regardless how efficient a company's operations process.
- On the *downstream side*, a company depends on its distributors and retailers for the delivery of the product to the final customer. If these are not delivered on time, are damaged in the transportation process, or are poorly displayed at the retail location, sales will suffer.
- If the operations function of other members of the supply chain is not managed properly, excess costs will result, which will be passed down to other members of the supply chain in the form of higher prices. The companies that comprise a supply chain need to coordinate and link their operations functions so that the entire chain is operating in a seamless and efficient manner.
- Just consider the fact that most of the components Dell uses are warehoused within a 15-minute radius of its assembly plant and Dell is in constant communications with its suppliers. Dell considers this essential to its ability to produce and deliver components quickly.



SECTION 2

PRODUCT DESIGN

PRODUCT DESIGN

- **Product design** is the process of specifying the exact features and characteristics of a company's product. Features of every product from a Starbucks cafe latte to Ikea's kitchen chairs to HP's DeskJet printer were determined during product design.
- All of a product's features are determined by product design. Ideas for which features to include are usually driven by marketing, which understands what customers want. Marketing understands that consumers respond to a product's appearance, color, texture, and performance. Marketing then works with operations to create a product that meets these consumer preferences.
- **Product design** decisions are one of the most important in OM as they drive many other decisions. One of these is **process design**, which is the design of the production process needed to produce the desired product. Product design defines a product's characteristics and then translates them into measurable dimensions the process can use to produce the product.
- **Product and process design** affect product quality, product cost, and customer satisfaction. Consider Apple's iPhone 4 manufactured in 2010. The product design specified white to be one of the color options. However, after white iPhones were produced, consumers found the white casing to 'leak' too much light into the phone unit, washing out pictures and creating other problems. If the product is not well designed, or if the production process is not capable of producing the exact product design features, the quality of the product will suffer. Therefore, for a product to succeed in the marketplace it must have a good design and a good production process.

PRODUCT DESIGN

DESIGN OF SERVICES

- *Product design* in the service industry has an added complication as the product is intangible and there is a high degree of customer interaction. *Service design* requires the design of the entire service concept. As with a tangible product, the service concept is based on satisfying customer needs.
- The service concept has *three elements*, each of which must be designed. The first is the *physical element* of the service. This may be a haircut at the barber shop, an investment made by the financial planner, or the filling your dentist just put on your tooth.
- The second are the *psychological benefits* of the service, which are elements such as promptness and friendliness.
- Finally, there are *sensual elements* of the service, which are part of the experience, such as ambiance, image, and 'feel good' elements of the service.
- Part of the service design is to specify all the elements of the service concept and to make sure that they work together. Consider the differences in service design between Hyatt Hotels, which provide upscale accommodations for travelers, and Hyatt Place, which provide lower priced accommodations but ample amenities for the business traveler.

PRODUCT DESIGN

PRODUCT DESIGN PROCESS

The product design process follows a predictable set of steps:

- *Idea development*
- *Product Screening*
- *Preliminary Design and Testing*
- *Final Design*

- 1. Idea development:** Product ideas can come from many sources, such as marketing, which tries to determine what customers want, a company's R&D, engineering, or suppliers. Often suppliers participate in a program called early supplier involvement (ESI) and get involved in the early stages of product design. Ideas can also come from competitors, by observing their products and success rate. Companies can also buy a competitor's product then analyze its parts and features. This is a process called reverse engineering, and was used by Ford in the design of its Ford Taurus.
- 2. Product Screening:** All product ideas go through a screening stage where they are evaluated to determine their likelihood of success. Product screening considers the viability of the product based on the needs of each of the business functions. Each function has its own concerns which are shown in Figure- 5. Deciding which ideas should pass the screening stage involves managerial skill and experience. Roughly 80% of ideas not making it past the screening stage due to the concerns illustrated in Figure-5. One decision making tool that can be especially helpful at this stage is break-even analysis.

PRODUCT DESIGN

- 3. Preliminary Design and Testing:** Following the screening stage a product idea moves into preliminary design and testing. This is the stage where design engineers translate vague and general performance specifications (e.g., ‘pretty blue color’ or ‘strong material’) into specific technical specifications. Proto types are built and tested, and changes are made. In service organizations this stage may entail pre-testing the product on small groups of customers and then refining the service concept accordingly.
- 4. Final Design:** Product testing results in the specification of the final product design. At this stage specifications are translated into specific instructions used to manufacture the product, such as materials, equipment, and quality standards.

Figure-5: Functional concerns at the product screening stage

Function	Product screening concerns
Marketing	What is the potential size of the market for the proposed product? How long will it take to capture the market? What is the long-term product potential?
Operations	What are the production process needs of the proposed product? Do we have the facilities, equipment, and labor skills needed? How easy would it be to acquire the needed production skills? Is there enough capacity to produce the new product?
Sourcing	What is the availability of sources of supply? What is the cost of materials for production? How easy is it to obtain available materials?
Finance	What are the financial requirements of the new product? What are the cost and the expected return on investment (ROI)

BREAK-EVEN ANALYSIS

- *Break-even analysis* is a tool that can be used to evaluate the success of a new product at the product screening stage.
- It is used to compute the product quantity that needs to be sold for a company to cover their costs, called the '*break-even point.*'
- Products with a high break-even point may be a risky choice as the company must sell a large quantity before it can cover costs.
- At the break-even point, *total cost equals revenue*, which is money made from sales.
- Total cost of producing a product or service is the sum of its *fixed and variable costs*. *Fixed costs* are those that do not vary with the quantity produced and include overhead, taxes, and insurance, and are incurred regardless of how much is produced.
- *Variable costs*, on the other hand, are costs that vary directly with the amount produced, and include items such as direct materials and labor.

BREAK-EVEN ANALYSIS

- Together, fixed and variable costs add up to total cost:

$$\text{Total Cost} = F + V_cQ$$

where,

F = fixed cost

V_c = variable cost per unit

Q = number of units sold

- Figure- 6 shows a graphical representation of these costs and the break even quantity.
- When $Q = 0$, total cost and fixed cost are equal. However, as Q increases, so do variable cost and total cost. Revenue from sales is computed as:

$$\text{Revenue} = S_P Q$$

where, S_P = selling price per unit

BREAK-EVEN ANALYSIS

- When $Q = 0$, revenue is zero, as there are no sales.
- As sales increase so does revenue, until the break-even point is reached when all costs are covered. Sales beyond the break-even point result in a profit for the company.
- At the break even point total cost is equal to revenue:

Total Cost = Revenue

$$F + V_c Q = S_p Q$$

Solving for Q at the break-even point— Q_{BE} —we get the following equation:

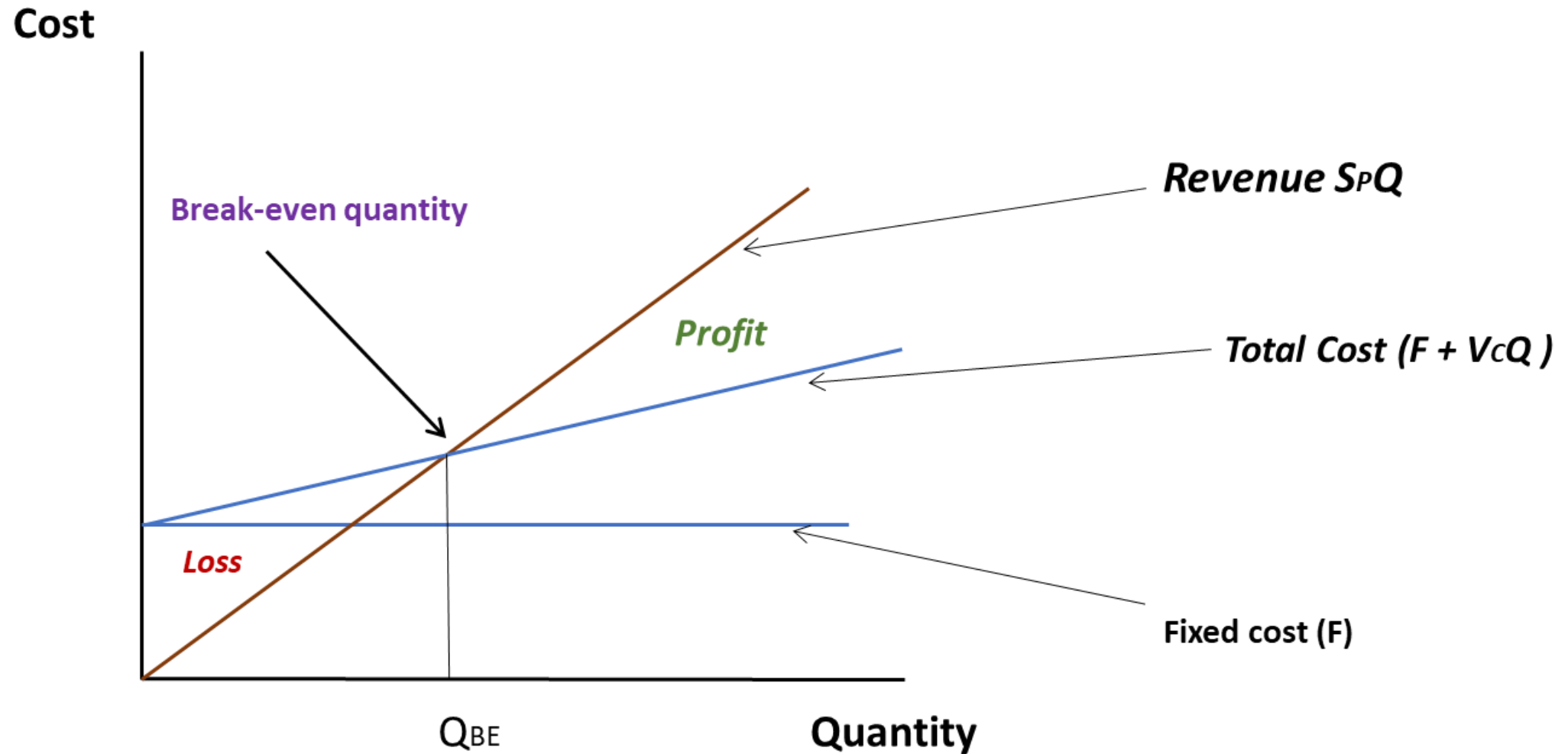
$$Q_{BE} = F / S_p - V_c$$

BREAK-EVEN ANALYSIS

- Q_{BE} computes the break-even quantity, or the quantity needed to be sold for the company to breakeven.
- The break-even point can also be arrived at by graphing total cost and revenue.
- Break-even analysis can be useful for conducting a sensitivity analysis to evaluate how changes in product price affect the break-even quantity.
- In addition, it can be used to make other product and process decisions, such as evaluating cost and benefits of different processes or deciding whether the company should make or buy a product.

BREAK-EVEN ANALYSIS

Figure- 6: Graphical representation of the break-even quantity



Example-1: Computing the Break-Even Quantity

Shu Chen plans to produce a new line of children's scarves and is considering the scarves' processing needs and market potential. She estimates that the variable cost for each product is \$25 and the fixed cost per year is \$15,000. If Shu offers each scarf at a selling price of \$30, how many scarves must she sell to break even? If Shu sells 4,000 scarves at the \$30 price, what will be the contribution to profit?

Solution: To compute the break-even quantity:

$$Q_{BE} = F / (S_P - V_C)$$

$$Q_{BE} = \$15,000 / \$30 - \$25 = 3,000 \text{ scarves}$$

This means Shu needs to sell 3,000 scarves to cover her costs. To compute the contribution to profit with sales of 4,000 scarves, we look at the relationship between cost and revenue:

$$\begin{aligned} \text{Profit} &= \text{Total Revenue} - \text{Total Cost} \\ &= (S_P)Q - [F + (V_C)Q] \end{aligned}$$

$$\begin{aligned} \text{Profit} &= \$30(4,000) - [\$15,000 + \$25(4,000)] \\ \text{Profit} &= \$5,000 \end{aligned}$$

The contribution to profit is \$5,000 if Shu can sell 4,000 scarves from her new line.

FACTORS IMPACTING PRODUCT DESIGN DECISIONS

There are some additional factors that need to be considered during the product design process:

- 1. Design for manufacture (DFM):** DFM is a series of guidelines that should be followed to produce a product easily and profitably. Consider that product design processes are typically driven by the desire to create the exact product the customer wants. However, it is also important to consider how easy it is to manufacture. Otherwise, a product that is either difficult or too costly to manufacture may be created. DFM guidelines focus on two issues: **design simplification and design standardization**. Design simplification means reducing the number of parts and features of the product whenever possible. **A simpler product** is easier to make, less mistakes are made, and less costly to produce. Design standardization, on the other hand, refers to the use of common and interchangeable parts. By using interchangeable parts, a greater variety of products can be made with less inventory, significantly lower cost, and greater flexibility. DFM includes rules such as minimize parts, design common parts for different products, use modular design, avoid tools, and simplify operations.

FACTORS IMPACTING PRODUCT DESIGN DECISIONS

2. Product Life Cycle:

- Most products go through a series of stages of changing product demand called the product life cycle.
- There are typically four stages: **introduction, growth, maturity and decline**.
- In the **introductory stage**, neither the product nor the market are well-defined, and all the “bugs” have not been worked out.
- In the **growth stage** the product is more established, although both product and market continue to be refined.
- In the **maturity stage**, the product and market become fairly stable and predictable.
- Finally, in the **decline stage** the product experiences declining demand due to competing product designs or market saturation.
- In the first two stages of the product life cycle called **the early stages** the product and market are still being refined.
- Here **product design** involves product improvement and refinement based on market response.
- In the last two stages called the **later stages** the product and market are stable.
- The product, however, is approaching the end of the life cycle and product design may involve major changes or reinventions to either revitalize an old product or create a new one for the market. .

FACTORS IMPACTING PRODUCT DESIGN DECISIONS

3. Concurrent Engineering:

- Historically, product design was made sequentially, where an idea was developed and then passed on to each function for evaluation.
- This was a time-consuming and inefficient process.
- Today, companies engage in concurrent engineering that requires all the functions to work together simultaneously to design and evaluate the product.

4. Remanufacturing:

- Remanufacturing uses components of old products in the production of new ones.
- It has enormous environmental and cost benefits, as the remanufactured products can be a fraction of the price of their new counterparts.
- Remanufacturing is especially popular in the electronics industry, such as the production of computers and televisions.

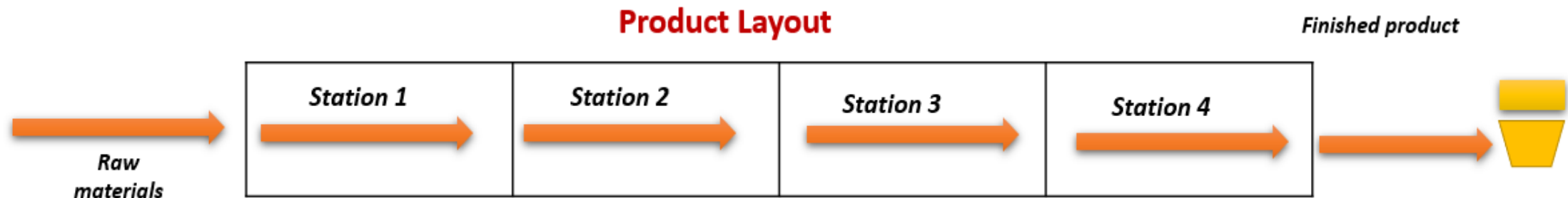
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SECTION 3

PRODUCT LAYOUT

Product Layouts:

- ✓ Arrange resources in sequence so the product can be made as efficiently as possible. This type of layout is used when producing a large volume of one standardized product and there is repetition of the process.
- ✓ It is a layout most suited for producing a high volume of one, or a few similar, products. An automotive assembly line, an automatic car wash, or a buffet table are perfect examples.
- ✓ Every product going through the line is almost exactly the same.



LINE BALANCING IN PRODUCT LAYOUTS

Line Balancing in Product Layouts:

Line balancing in a **product layout** is the process of assigning tasks to workstations to equalize workload, minimizing idle time, and maximizing efficiency, often in a continuous flow.

Line balancing involves the following actions

- The logical order of task to be performed
- Time required to perform each task.
- The speed of the production process (time specification to perform the assigned task).

Steps in designing product layouts

Step 1: Identify Task Times and Precedence Relationships

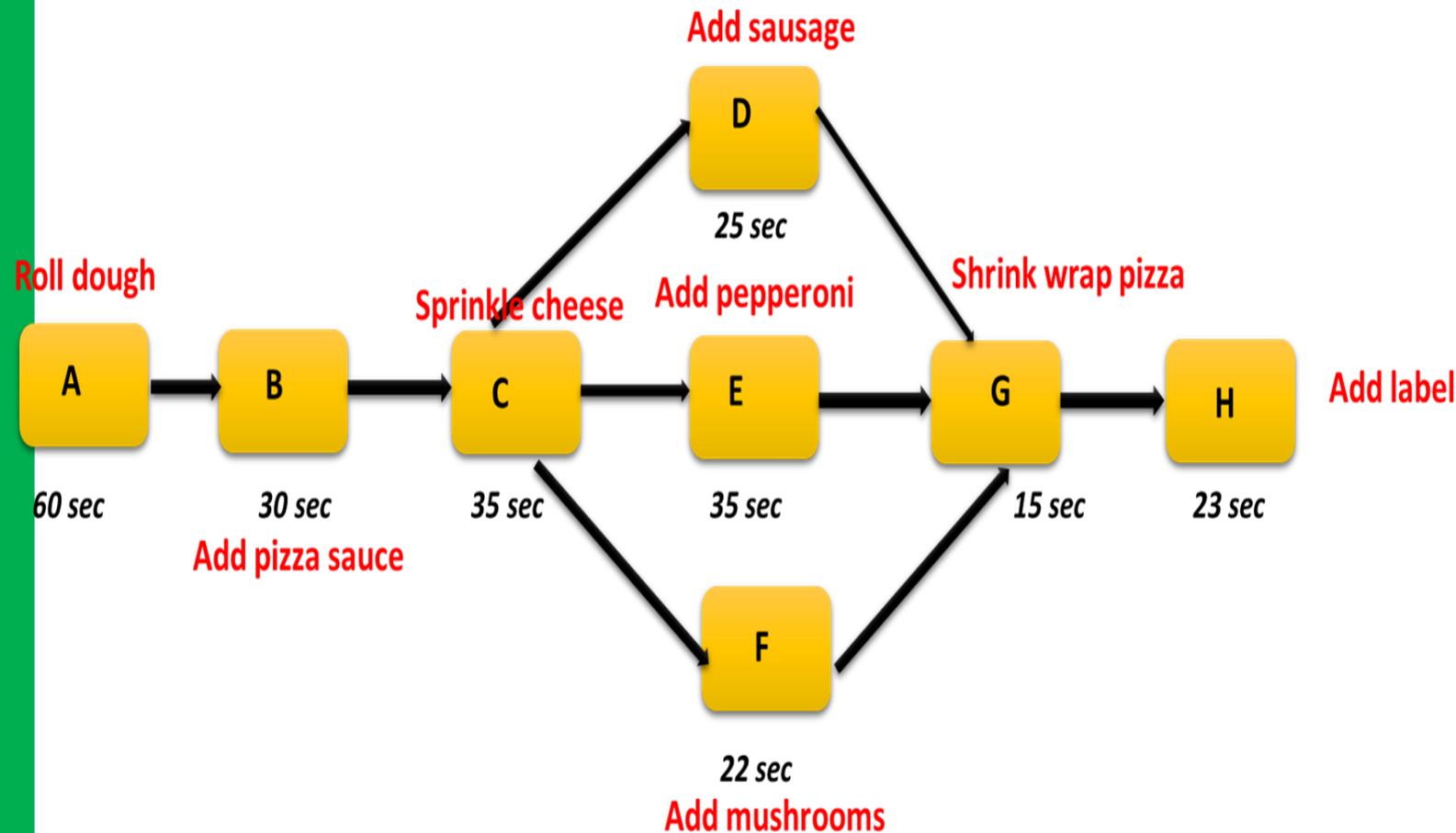
In line balancing the first step involves the identification of product to be produced, their time duration, and the sequence need to be followed. For example, the task of a *pizza assembly*.

Figure : Precedence relationships for pizza assembly

Task	Description	Task time (sec)	Preceding task
A	Roll dough	60	-
B	Add pizza sauce	30	A
C	Sprinkle cheese	35	B
D	Add sausage	25	C
E	Add pepperoni	35	C
F	Add mushrooms	22	C
G	Shrink wrap pizza	15	D, E, F
H	Add label	23	G
Total time		245 sec	

Steps in designing product layouts

Figure : Precedence diagram of pizza assembly



Task	Description	Task time (sec)	Preceding task
A	dough	60	-
B	pizza sauce	30	A
C	Sprinkle cheese	35	B
D	sausage	25	C
E	pepperoni	35	C
F	mushrooms	22	C
G	Shrink wrap pizza	15	D, E, F
H	label	23	G
Total time		245 sec	

- Notice that in *pizza assembly* the dough must be rolled before the sauce can be added. However, pepperoni, or mushrooms are added only after the cheese.

Steps in designing product layouts

Step 2: Determine Cycle Time

- **Cycle time** or **takt time** refers to the maximum amount of time each station on the assembly line takes to complete the assigned task.
- The actual work time assigned to each station cannot exceed the assigned time, or cycle time. Otherwise there would not be enough time to complete the work.
- This indicates cycle time is directly related to the number of unit produced.
- The total task time to produce **one pizza is 245 second (step 1)**, if a single person singlehandedly carries out all the tasks from beginning to end without a break, the maximum output in one hour would be:

$$\text{Maximum output} = \frac{3,600 \text{ seconds/hour}}{245 \text{ seconds/hour}} = 14.7 \text{ pizzas/hour}$$

- However, if we want to produce more pizzas than **14.7 pizzas/hour**, we have to divide the work among a number of people working simultaneously at workstations to achieve the desired output rate. The first thing to compute is the cycle time:

Cycle time = Amount of time allowed to complete work at each station

$$\text{Cycle time} = \frac{\text{Available production time per day}}{\text{Desired number of units per day}}$$

- Note that the time period are to compute cycle time are per day, but they can also be given per hour or another time period, as long as the time periods are consistent. Here, the operations run continuously for **eight hours** a day. Example, we want to produce **300 pizzas** per day.

$$\text{Cycle time} = \frac{28,800 \text{ seconds per day}}{300 \text{ pizzas per day}} = 96 \text{ seconds per workstation}$$

Steps in designing product layouts

Step 3: Determine the Theoretical Minimum Number of Workstations

- The goal in line balancing is to assign tasks to workstations to achieve the best efficiency.
- This usually helps to achieve a baseline of what is theoretically the minimum possible number of stations we need, although the line balancing may have more stations.
- The theoretical minimum number of stations computes the number of stations needed if we are 100% efficient.

$$\text{Theoretical minimum no. of stations (N)} = \frac{\text{Sum of the task times}}{\text{Cycle time}}$$

$$N = \frac{245 \text{ seconds}}{96 \text{ seconds per station}} = 2.55 \text{ stations} = 3 \text{ stations}$$

- Note that when determining the *number of stations*, round up the next whole number, as we can never have a *partial station*.

Steps in designing product layouts

Step 4: Assign Tasks to Workstations

We can now proceed to assign tasks to workstations. Note that we have to follow our precedence relationships to ensure the right sequence. For example, this means that you cannot put the cheese on before adding the sauce.

Figure : Assigning tasks to workstations

WORK STATION	ELIGIBLE TASK	ASSIGNED TASK	TIME REQUIRED	CUMULATIVE TIME REQUIRED	IDLE TIME
1	A	A	60S	60S	$96S - 60S = 36S$
1	B	B	30S	90S	$96S - 90S = 6S$
2	C	C	35S	35S	$96S - 35S = 61S$
2	D, E, F	E	35S	70S	$96S - 70S = 26S$
2		D	25S	95S	$96S - 95S = 1S$
3	F	F	22S	22S	$96S - 22S = 74S$
3	G	G	15S	37S	$96S - 37S = 59S$
3	H	H	23S	60S	$96S - 60S = 36S$

Selection of task is made using the longest task time rule.

There is not enough time remaining to add more tasks to his station.
Start another station.

As we assign tasks to workstations, following the precedence, we have to make sure that the sum of the task times at any station does not exceed the cycle time. Otherwise, the person working at that station will not have time to complete his or her task. Also, sometimes we have a choice of tasks to assign to a station. For example, we can decide whether to put either pepperoni or mushrooms or sausage next in the sequence.

Steps in designing product layouts

There are two rules we can follow here to make it easier:

- Select the task with the longest tasks time (**Longest Task Time Rule**)
 - Select the task with most number of followers (**Number of Followers Rule**)
-
- These are just rules of thumb, or what we call “**heuristics.**”
 - They will not necessarily result in the optimal solution, but will give us a good solution.
 - In our example the **longest task time rule** is used.
 - However, when assigning tasks in large and complex processes multiple rules are used to balance the line, all with the goal of providing the highest efficiency.

Steps in designing product layouts

Step 5: Compute Efficiency

Once we have assigned tasks to workstations on the line we need to compute our efficiency:

$$\begin{aligned}\text{Efficiency} &= \frac{\text{Sum of task times}}{\text{Number of workstations} * \text{Cycle time}} \\ &= \frac{245 \text{ seconds}}{3 \text{ stations} * 96 \text{ seconds /station}} \\ &= 0.8507 \approx 85.07\%\end{aligned}$$

The efficiency could be improved with better utilization of the line.

- One option may be to split tasks into smaller work elements or change technology to reduce the time required.
- More workers can also be assigned to the bottleneck tasks, which is **task A (Roll dough) in our example.**

This could reduce the time it takes and allow us to balance the line more efficiently.